FloatChat: AI-Powered Ocean Data Exploration

A natural language interface that transforms how scientists interact with ARGO ocean data, making complex scientific datasets intuitive and interactive through conversational AI.

TEAM LEADER : HARISH V K
TEAM LEADER MAIL ID: harishkumarvk05@gmail.com
TEAM NAME: Git happens
TEAM MEMBERS: HARISH L, GUNASAKTHI L

PROBLEM STATEMENT :-
ID: PS 1
STATEMENT: FloatChat - AI-Powered Conversational Interface for ARGO Ocean Data Discovery and Visualization

The Challenge with ARGO Data

Current Format Limitations

ARGO data is stored in NetCDF format—a highly structured but complex scientific file format that resists real-time querying and conversational interfaces.

The multidimensional structure makes data interpretation challenging without specialized knowledge and tools.

Not Real-Time
Difficult to query on demand

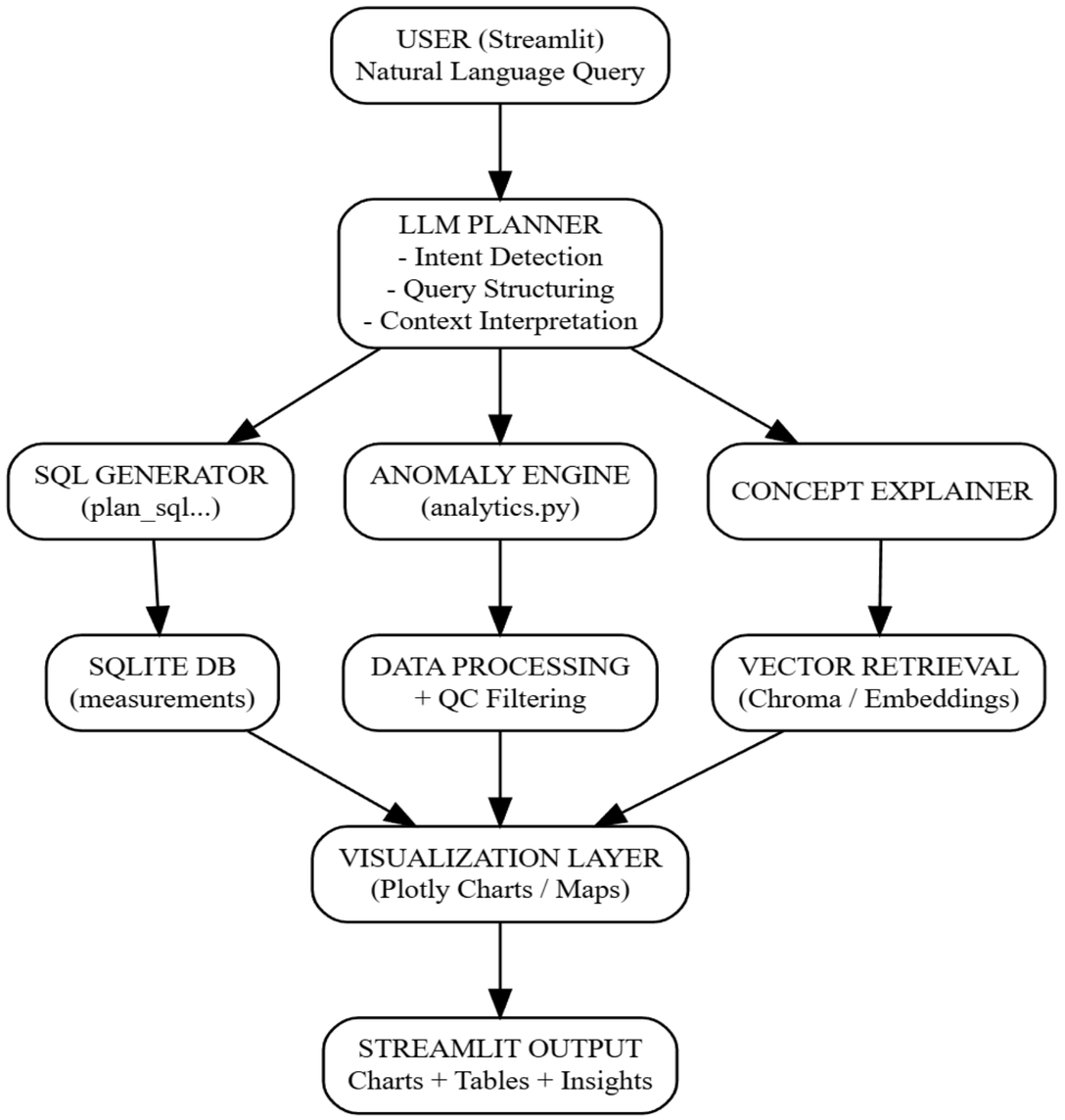
Not Conversational
Requires technical expertise

Complex Structure
Hard to interpret patterns

Goal: Enable intuitive, natural language exploration of oceanographic data

System Overview: End-to-End Architecture

A full-stack intelligent data system that bridges raw ocean measurements with human-friendly interaction.



Data Engineering Pipeline

01

Parse NetCDF Files

Extract raw measurements from scientific format

02

Standardize Variables

Normalize temperature, salinity, and pressure readings

03

Convert to Parquet

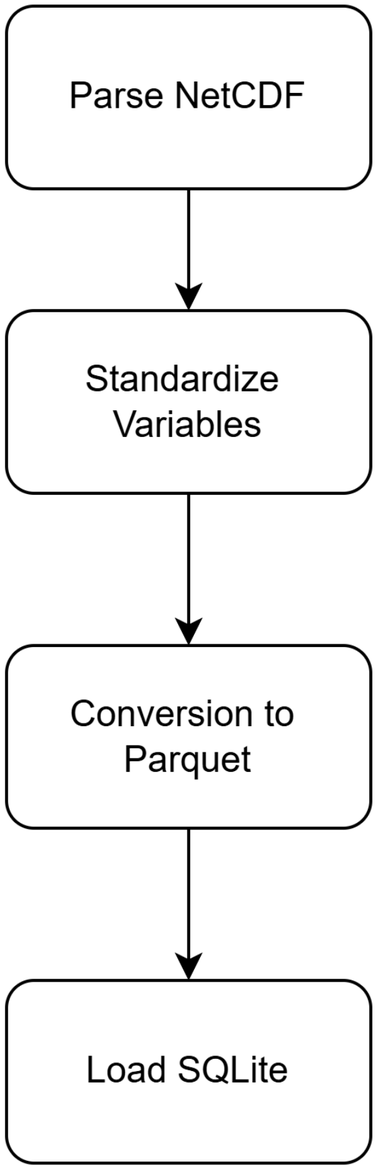
Optimize for fast columnar queries

04

Load SQLite

Enable structured, indexed access

This pipeline transforms complex multidimensional data into a query-optimized structure that supports real-time analysis and visualization.



The Core Challenge: Independent Rows

Problem with Current Structure

Data stored as independent rows with no natural grouping mechanism. Measurements from a single float profile are scattered across thousands of entries.

> How do you analyze vertical ocean behavior when measurements aren't logically grouped?
>
> Without profile context, tracking float measurements through depth becomes nearly impossible.

float_id	timestamp	pressure (dbar)	temperature (°C)	salinity (PSU)
5901234	2024-01-10 10:00	10	24.52	35.12
5901234	2024-01-10 10:00	20	23.87	35.05
5901234	2024-01-10 10:00	30	22.94	34.97
5901234	2024-01-10 10:00	40	21.73	34.89

Table: Raw ARGO Measurements (Before Structuring)

Innovation: Profile ID System

Profile Metadata

profile_id	float_id	cycle	timestamp	direction	depth_range	points
5901234_012_20240110_A	5901234	12	2024-01-10 10:00	Upward	10–40 m	4

Profile Measurements

profile_id	pressure (dbar)	temperature (°C)	salinity (PSU)
5901234_012_20240110_A	10	24.52	35.12
5901234_012_20240110_A	20	23.87	35.05
5901234_012_20240110_A	30	22.94	34.97
5901234_012_20240110_A	40	21.73	34.89

profile_id = float_id + cycle + timestamp + direction

Smart Grouping Key

profile_id combines float_id, cycle number, timestamp, and measurement direction

Vertical Profiles

Groups all measurements from a single descent or ascent cycle

Meaningful Analysis

Enables depth-based visualization and behavioral tracking

This innovation transforms independent measurements into coherent vertical profiles—unlocking scientific analysis that was previously impossible.

Anomaly Detection Evolution

Initial Approach: Global IQR

Used **Interquartile Range** with **fixed thresholds** across all depths and locations.

Why it failed: Ocean properties vary dramatically with depth, latitude, and season. Global thresholds **flagged valid measurements as anomalies.**

Improved Strategy: Contextual Detection

Grouped measurements by depth zones and float behavior patterns. Applied **robust Z-score** within context.

Result: **Accurate anomaly detection** that respects oceanographic variability.

IQR Failure (Anomaly Problem)

pressure	temperature	IQR Range (Global)	Result
10	24.52	22–26	Normal
20	23.87	22–26	Normal
30	22.94	22–26	Normal
40	21.73	22–26	Anomaly

Improved Strategy (Context-Aware Detection)

depth_band	temperature	mean	std_dev	z-score	anomaly
0–20 m	24.52	24.3	0.4	0.55	NO
20–40 m	21.73	22.1	0.3	-1.23	NO
20–40 m	20.10	22.1	0.3	-6.7	YES

Quality Control Integration

1. **QC Flag Analysis**

   Parse quality control flags embedded in ARGO data

2. **Data Filtering**

   Remove unreliable measurements before anomaly detection

3. **Context-Aware Detection**

   Apply statistical methods only to trusted data

By combining automated QC flags with contextual anomaly detection, we ensure scientifically valid results that oceanographers can trust.

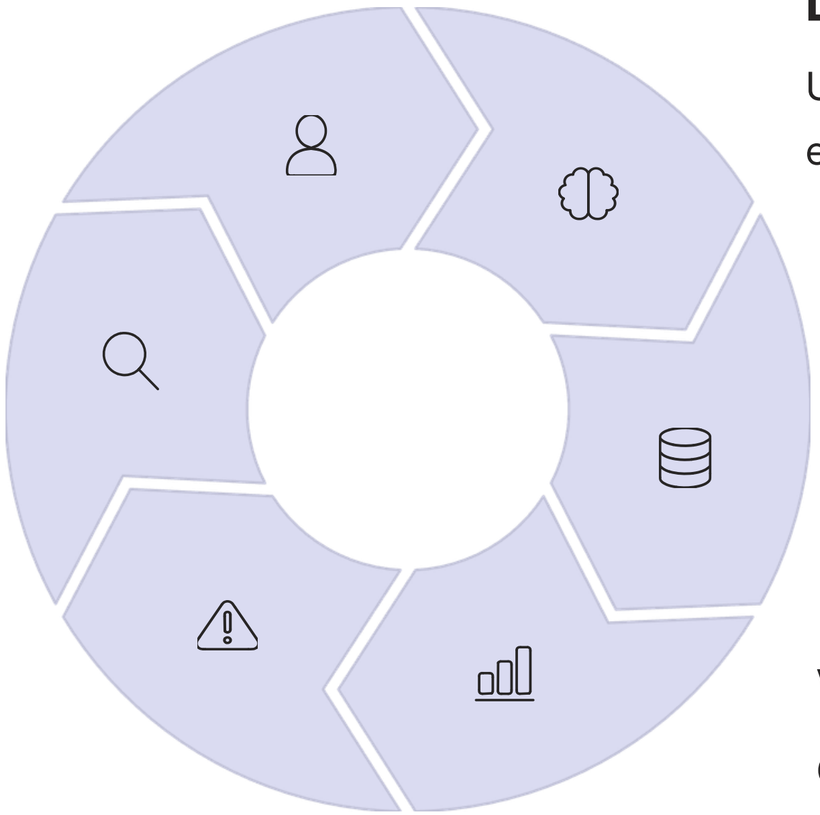
AI Architecture: The Planner System

User Query

Natural language question

LLM Planner

Understands intent, generates execution plan



Vector Retrieval

Grounds queries in schema

SQL Queries

Retrieves relevant data

Anomaly Engine

Detects unusual patterns

Visualization

Generates plots

The planner acts as an intelligent orchestrator—breaking complex questions into executable steps while vector retrieval ensures accurate schema understanding.

Conclusion: Bridging Data and Understanding

Profile-Based Structuring

Context-aware grouping enables vertical ocean analysis

Context-Aware Anomalies

Depth and location-sensitive detection

Planner-Based AI

Explainable, accurate responses

FloatChat transforms raw oceanographic data into an intuitive, conversational interface—empowering scientists to explore ARGO data through natural language queries.

By addressing the fundamental challenges of data structure, anomaly detection, and AI planning, we've created an intelligent system for real-time ocean data exploration.